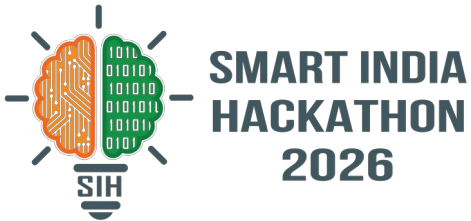


SMART INDIA HACKATHON 2026



RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

SIH26002

MDoNER • Smart Automation • Software

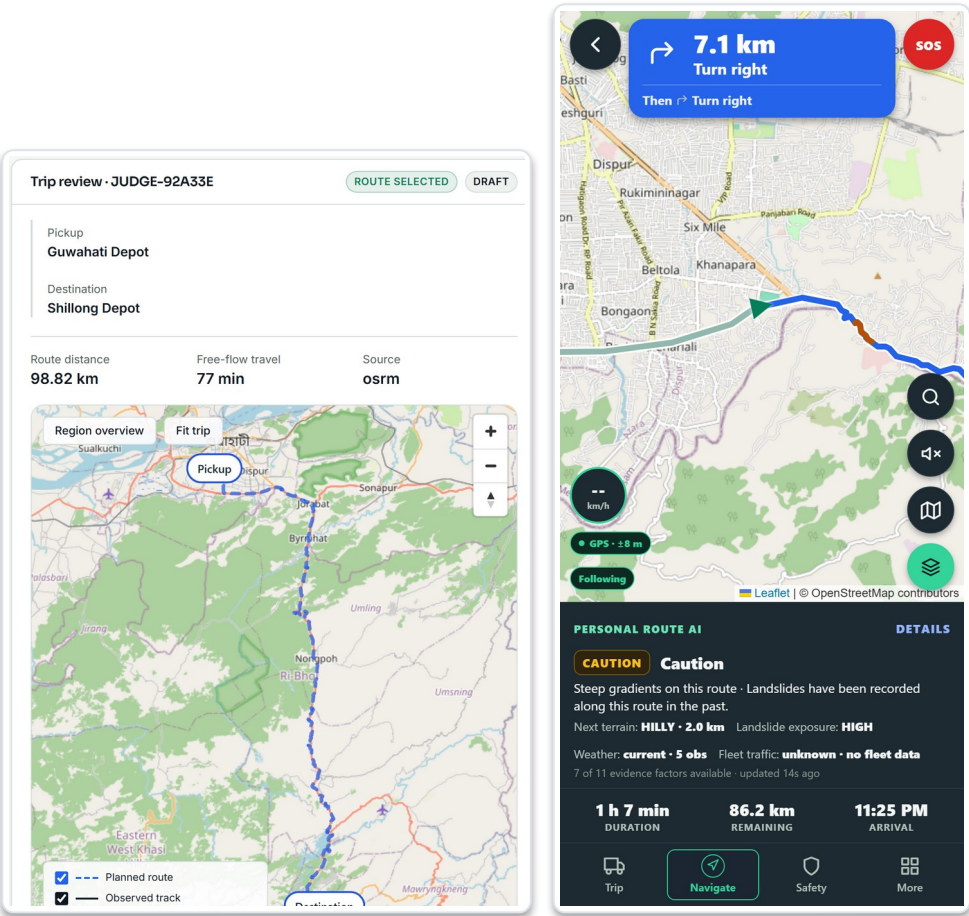
PROBLEM STATEMENT ID	SIH26002
PROBLEM STATEMENT TITLE	AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)
THEME	Smart Automation
PS CATEGORY	Software
TEAM ID	17 (internal group no.)
TEAM NAME	NER-AI LOGISTICS

SHORTEST IS NOT ALWAYS USABLE.

“Can this essential-supply truck reliably use this corridor now?”

End-to-end working prototype • physical-phone validated • live demo corridor Guwahati → Shillong

LIVE WORKING SYSTEM



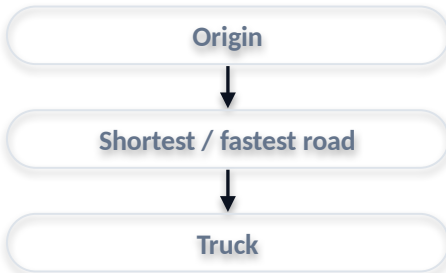
Manager route review • Driver navigation — real screens

PROPOSED SOLUTION

Not the shortest road. **The road that is operationally usable now.**

NER corridors close without notice: monsoon rain, landslides, floods, official warnings, single-lane hill roads, no phone signal. A shortest-path router still sends the truck.

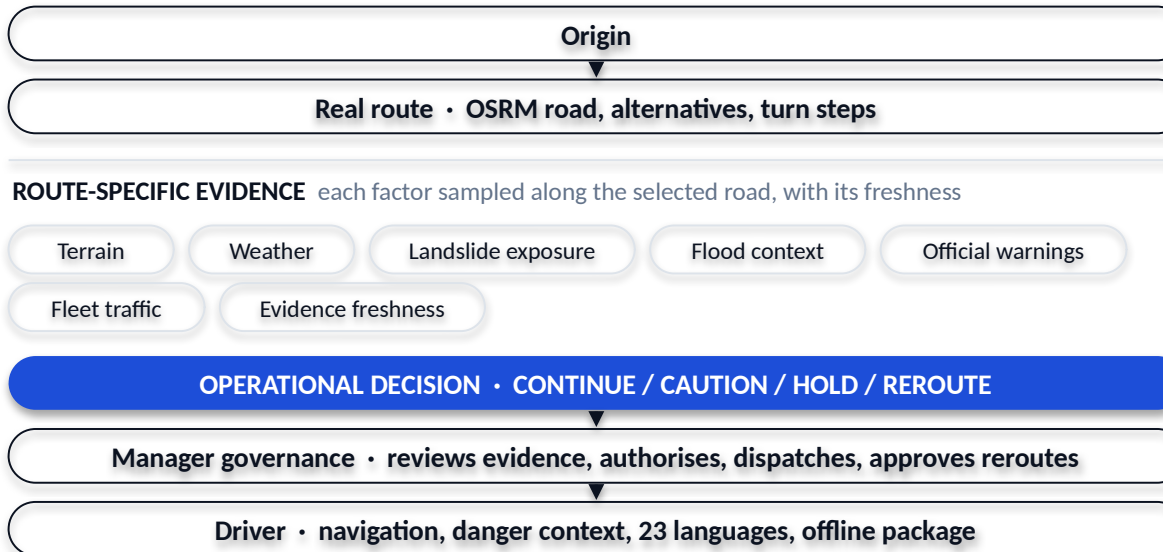
A NORMAL ROUTER



No terrain, weather, hazard or freshness check.

No governance. No answer when data is missing.

RASTA AI



UNKNOWN ≠ SAFE

Missing or stale evidence is marked UNKNOWN, never counted as safe, and the route goes to an authorised reviewer.

MANAGER • CHECK CONDITIONS

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN
0.5 KM STEEP

49-1436 m · 1975 m climb · steepest 10.3%
flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY
HIGH EXPOSURE

22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog · inventory 2007-2017 — aged, recent years not covered

RIVER LEVELS
NORMAL

Discharge at 5 river cells along the corridor, highest 1.0x its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS
NONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC
UNKNOWN

Real screen · evidence · freshness · UNKNOWN

TECHNICAL APPROACH

VERIFIED DATA → DECISION → DRIVER

WEATHER Open-Meteo · MET Norway	TERRAIN Copernicus DEM · OpenTopoData	WARNINGS NDMA SACHET (CAP)	FLOOD GloFAS	LANDSLIDE HISTORY NASA GLC 2007–17	TRAFFIC fleet GPS fixes
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ROUTE-SPECIFIC EVIDENCE · FastAPI backend

every factor sampled along the selected road · PostgreSQL / PostGIS (Supabase) · OSRM routes, alternatives, turn steps

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE**CAUTION****HOLD****REROUTE****MANAGER REVIEW****DRIVER NAVIGATION****ROUTE-AHEAD MONITOR (60 s)****CONDITIONS CHANGE?****REASSESS / REROUTE**

Conditions are re-scored every 60 s along the road ahead; a material change goes back through the policy and a reroute is proposed to the manager — never applied silently.

CLIENTS Manager console — React + TypeScript, manager accounts only · Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package

STACK FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

EVIDENCE Open-Meteo / MET Norway · NDMA SACHET · GloFAS · NASA historical landslides · OpenTopoData / Copernicus DEM

WHO DECIDES

DETERMINISTIC SAFETY ENGINE
makes every operational decision

GEMINI / OPENROUTER (LLM)
explains the decision in the driver's language — never decides

EXPERIMENTAL ML
research channel only; held outside routing by the safety gate

GOVERNANCE CHAIN

Verified evidence**Deterministic safety policy****Manager governance****Driver action**

Not a concept. A working end-to-end prototype.

PLAN

REVIEW

DISPATCH

ACCEPT

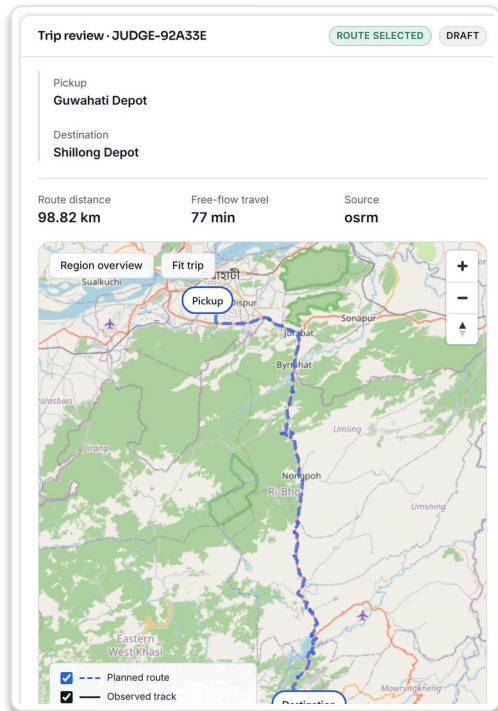
TRUCK VERIFY

NAVIGATE

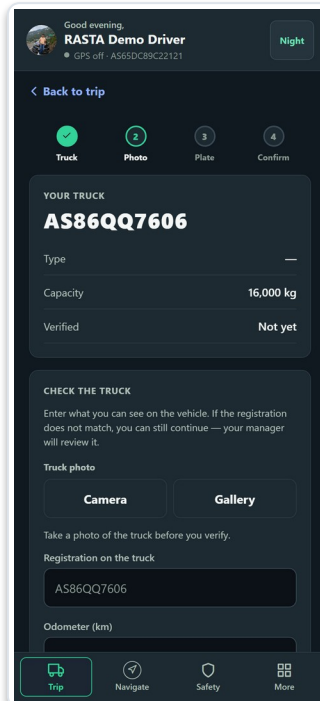
OFF-ROUTE

MANAGER REROUTE APPROVAL

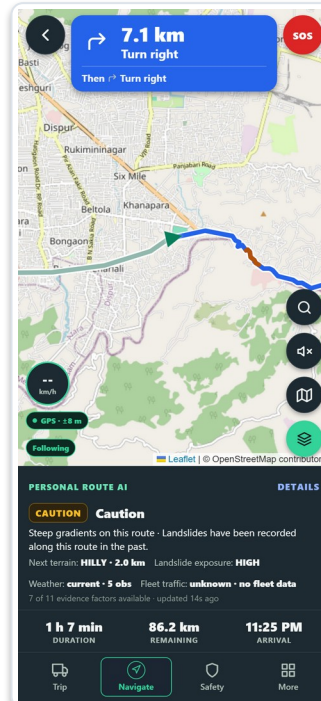
DELIVERY



Manager · route review



Driver · truck check



Driver · navigation

PROOF

The whole lifecycle above was run end to end on a physical Android phone against the hosted backend, on the real road.

12/12 physical judge-flow steps

9/9 physical role-flow steps

1138+ backend · 621+ driver · 170+ manager tests

PHYSICAL ANDROID · CERTIFIED

Canonical demo Guwahati → Shillong · ≈ 98.8 km real route

KNOWN LIMITS → MITIGATION Render cold start → warm /health before the demo · provider rate limits → cached evidence + explicit UNKNOWN · background push needs Firebase config → in-app alerts already work · experimental ML → outside routing until the gate passes

IMPACT AND BENEFITS

RASTA AI does not ask only “Which road is shortest?” It asks “Can this truck reliably use this corridor now?”

FOOD • MEDICINE • FUEL • RELIEF SUPPLIES — essential logistics for hill communities, decided on evidence

WHO BENEFITS

FLEET MANAGERS

Route usability and uncertainty visible before dispatch; auditable decisions.

DRIVERS

Safer navigation, danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

More resilient access to food, medicine, fuel and relief supplies.

GOVERNMENT / OPERATIONS

Auditable route evidence and disruption visibility per corridor.

DEMONSTRATED TODAY

23 selectable languages, explicit fallback status

11 route factors in every decision

60 s route-ahead monitor while driving

12/12 judge-flow steps certified on a physical phone

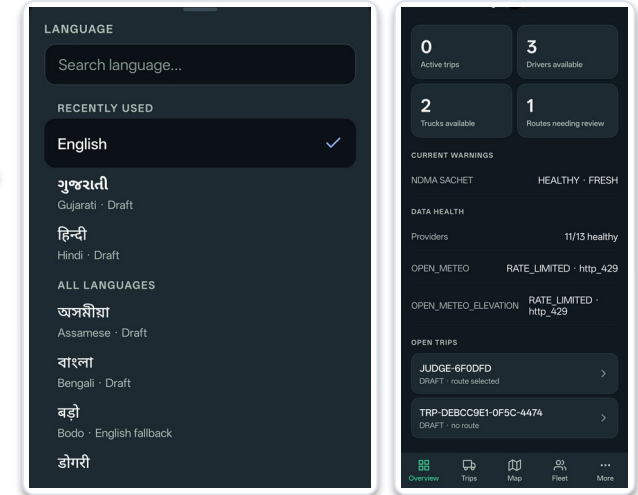
100 % of reroutes reviewed by a human

0 unknown factors counted as safe

HUMAN GOVERNANCE — AI HAS NO UNCONTROLLED AUTHORITY



REAL SCREENS • APK 1.0.18



Driver • 23 languages

Manager • mobile

Real screens, APK 1.0.18. One login, server-decided role: a manager gets a mobile fleet view, a driver gets navigation. Each language shows its status — Verified, Draft or English fallback — so nobody is misled.

RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

ROUTING / PLACES

OSRM • OpenStreetMap • Nominatim

real road geometry, alternatives and turn steps; place search on the corridor

OFFICIAL / FLOOD

NDMA SACHET (CAP) • Copernicus GloFAS

official warnings polled with freshness; river discharge against its 30-day mean

AI ASSISTANCE

Google Gemini

explains the decision in the driver's language — never safety authority

WEATHER / TERRAIN

Open-Meteo • MET Norway • Copernicus DEM •**OpenTopoData**

forecast sampled along the road; climb, gradient and steep segments (GLO-90)

LANDSLIDE HISTORY

NASA Global Landslide Catalog

historical exposure; inventory 2007–2017, ≤5 km location-accuracy filter

PROBLEM STATEMENT

MDoNER • SIH26002

AI-based smart logistics and accessibility intelligence for the NER

EXPERIMENTAL LANDSLIDE RESEARCH

95 %**RECALL**

on the NER geographic holdout — 154 catalogued events, no NER row used in training

SAFETY GATE

NOT DEPLOYED**FPR 50 %** too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. Strong recall was not enough. Held-out testing exposed the false-positive rate, so the gate keeps the model out of routing.

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 • calibrated logistic regression on NASA GLC + ERA5-Land rain + DEM slope • docs/HAZARD_ERROR_ANALYSIS.md

REFERENCES

gpm.nasa.gov/landslides (NASA GLC) • sachet.ndma.gov.in • global-flood.emergency.copernicus.eu (GloFAS) • open-meteo.com • api.met.no • opentopodata.org • Copernicus DEM • project-osrm.org • openstreetmap.org • github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry)